ENSEMBLE CLASSIFIERS

From Bagging & Boosting to Validation, Imbalanced Data & ROC Curves

BAGGING

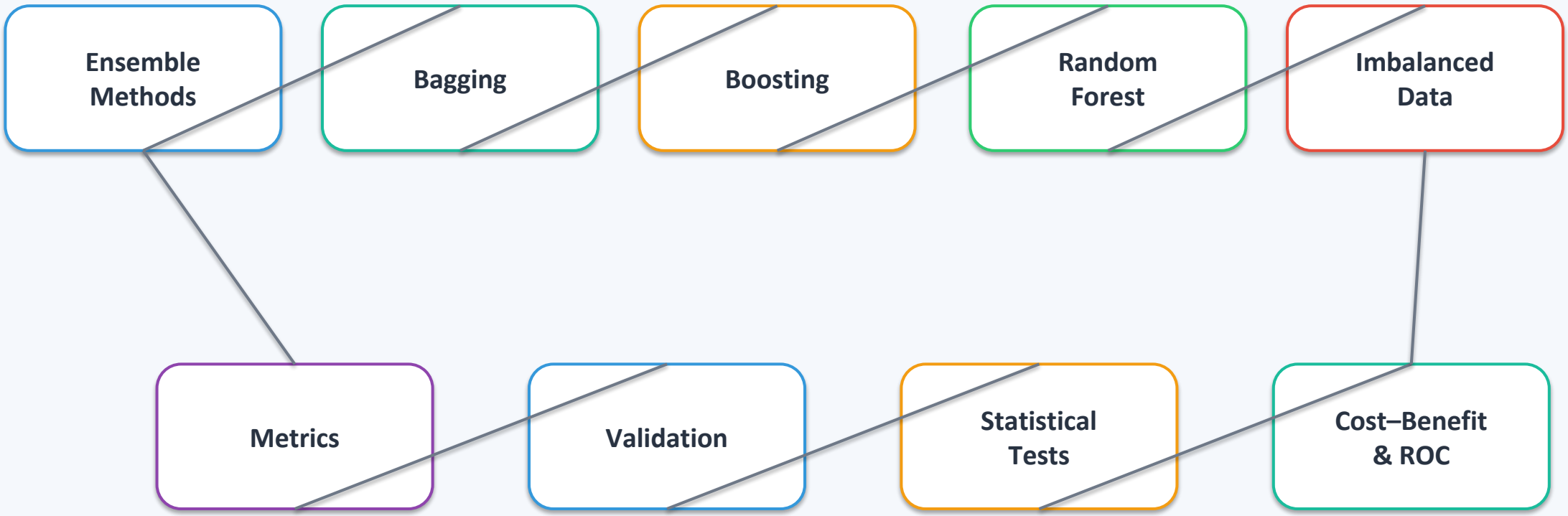
BOOSTING

RANDOM FOREST

EVALUATION

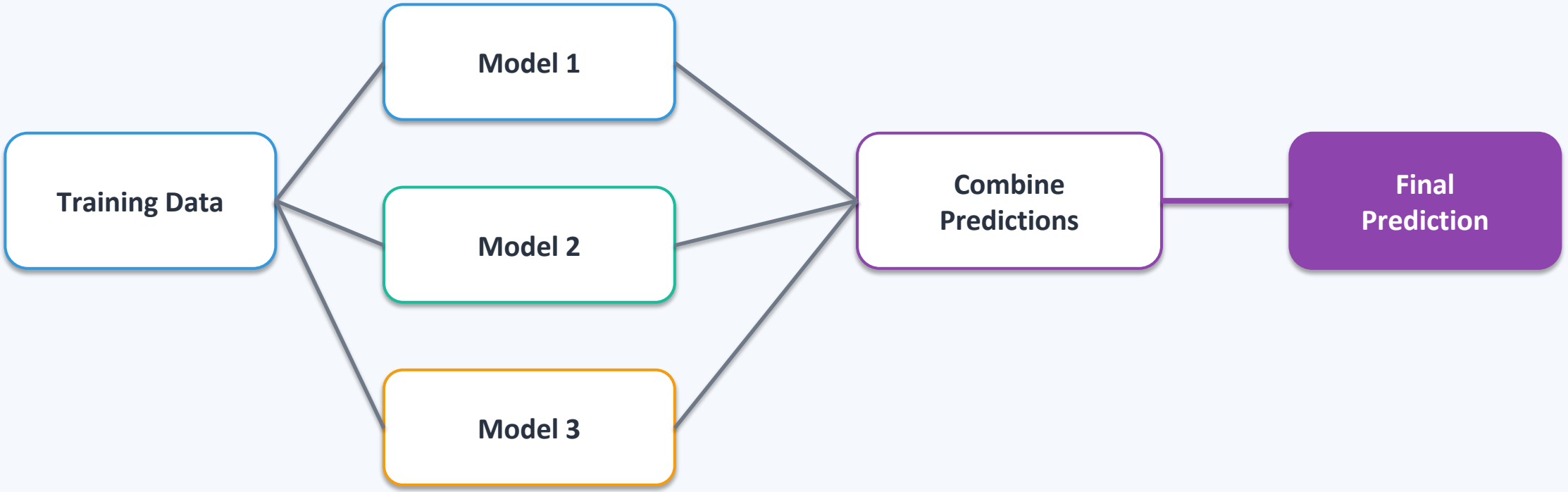
Learning Roadmap

A visual map of the complete topic



1. Introduction to Ensemble Methods

Combine several models to obtain a stronger final predictor



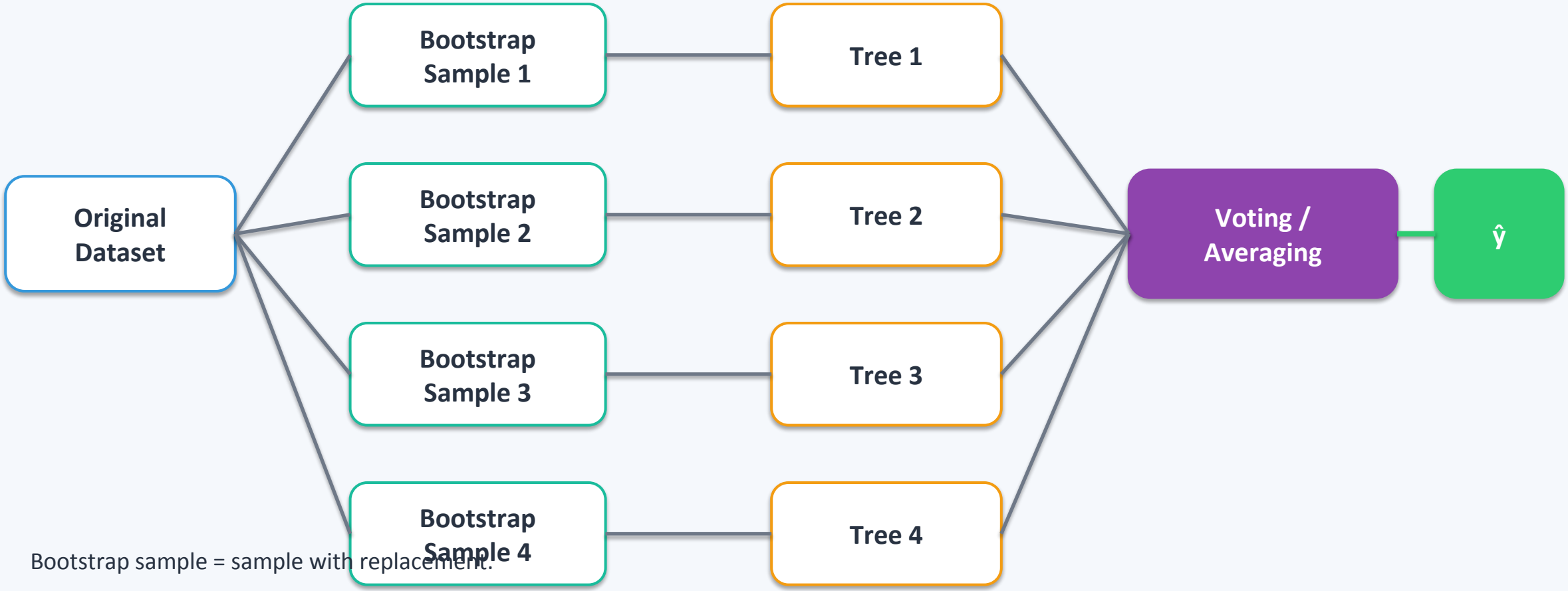
Key idea: many diverse models can be better than one model.

Diversity matters: models should make different errors.

Common approaches: Bagging, Boosting and Random Forests.

2. Bagging — Bootstrap Aggregating

Train models independently on different bootstrap samples



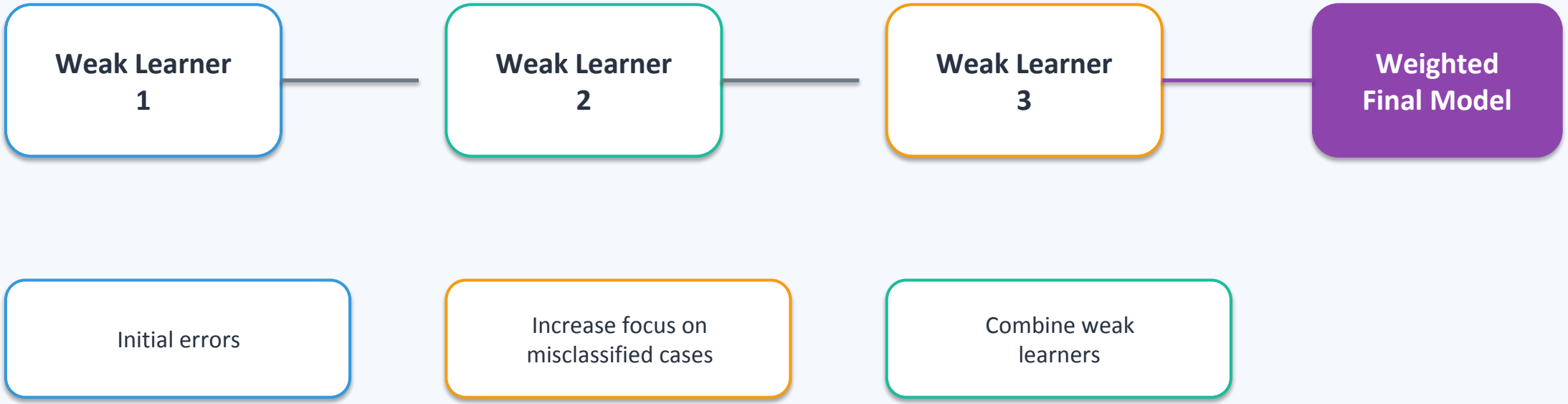
Bootstrap sample = sample with replacement.

Models train in parallel → reduces variance.

Bagging is especially effective for unstable learners such as decision trees.

3. Boosting

Build models sequentially; later models focus on previous errors



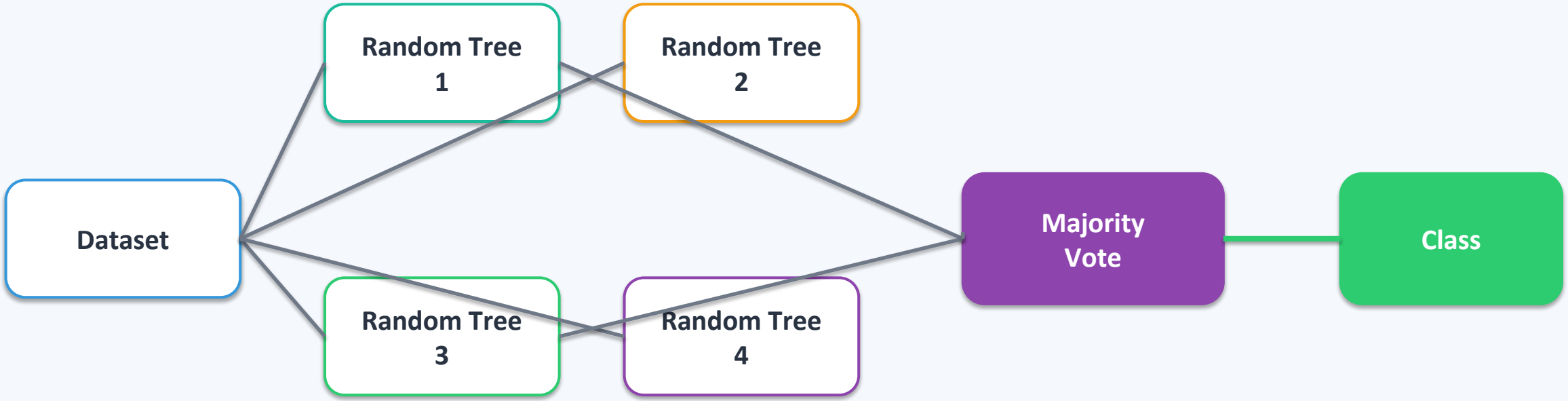
AdaBoost: increases weight of incorrectly classified observations.

Gradient Boosting: each learner learns to reduce the current loss.

Strength: can produce highly accurate models; risk: overfitting if poorly tuned.

4. Random Forest

An ensemble of decision trees using bootstrap samples + random feature selection

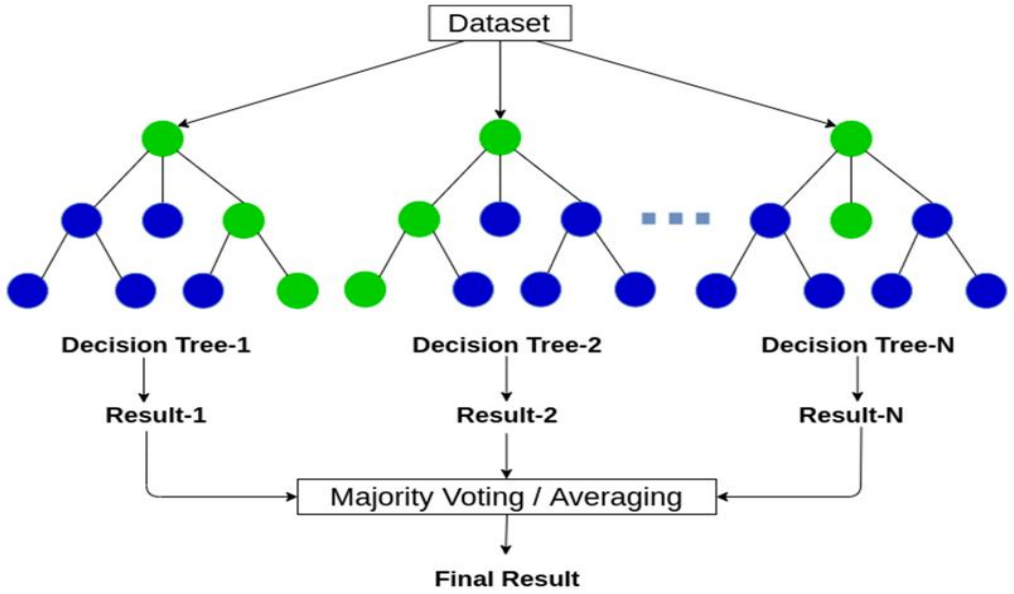


Each tree sees a bootstrap sample.

At each split, only a random subset of features is considered.

Randomness decorrelates trees → strong generalization and lower variance.

Random Forest



- Reduces overfitting compared with a single decision tree.
- Works well with large datasets.
- Can handle many features.
- Useful for both classification and regression.
- Does not require extensive feature scaling.
- Generally provides good predictive performance.
- Can provide an estimate of feature importance.

5. Improving Classification Accuracy of Class-Imbalanced Data

Accuracy alone can hide poor performance on the minority class



Oversampling: duplicate or synthesize minority examples.

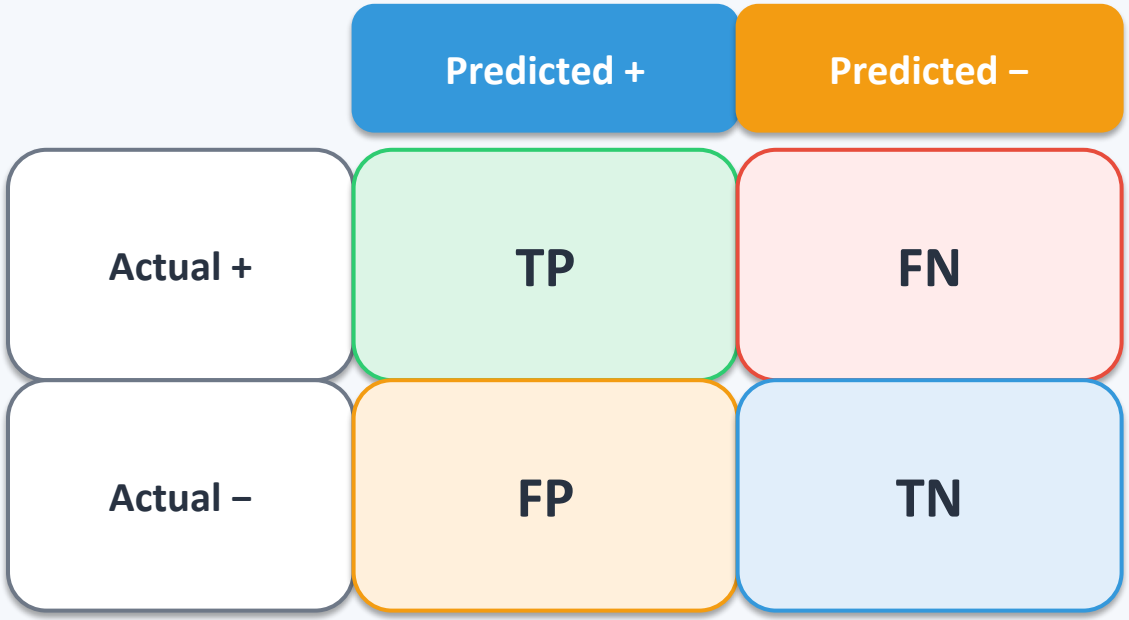
Undersampling: reduce majority examples.

SMOTE: create synthetic minority examples.

Use precision, recall, F1, balanced accuracy, PR-AUC—not accuracy alone.

6. Metrics for Evaluating Classifier Performance

Start with the confusion matrix



Accuracy

(TP+TN)/Total

Overall correctness

Precision

TP/(TP+FP)

How many predicted + are correct

Recall

TP/(TP+FN)

How many actual + are found

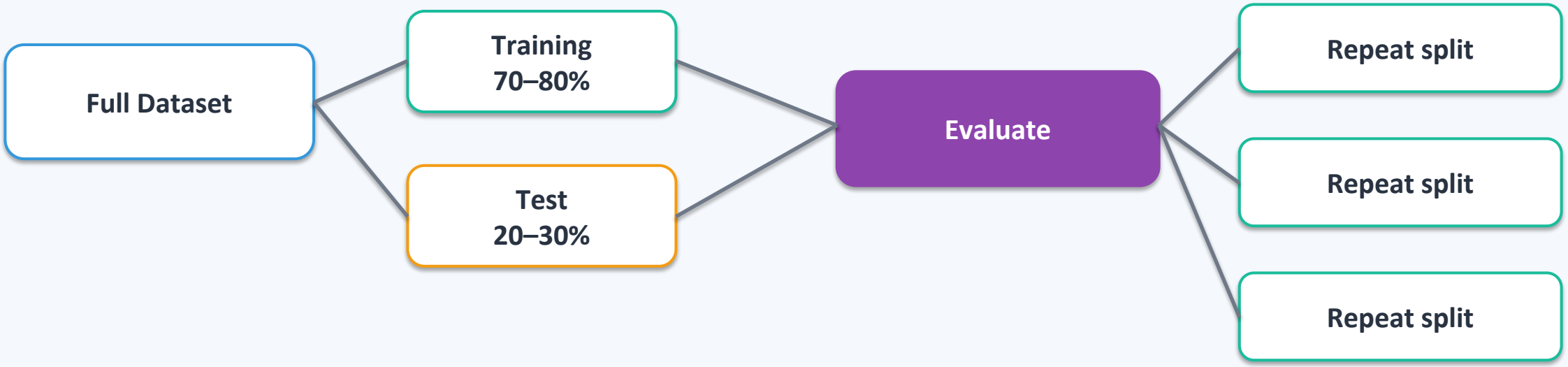
F1

2PR/(P+R)

Balance of precision & recall

7. Holdout Method & Random Subsampling

Simple ways to estimate generalization performance



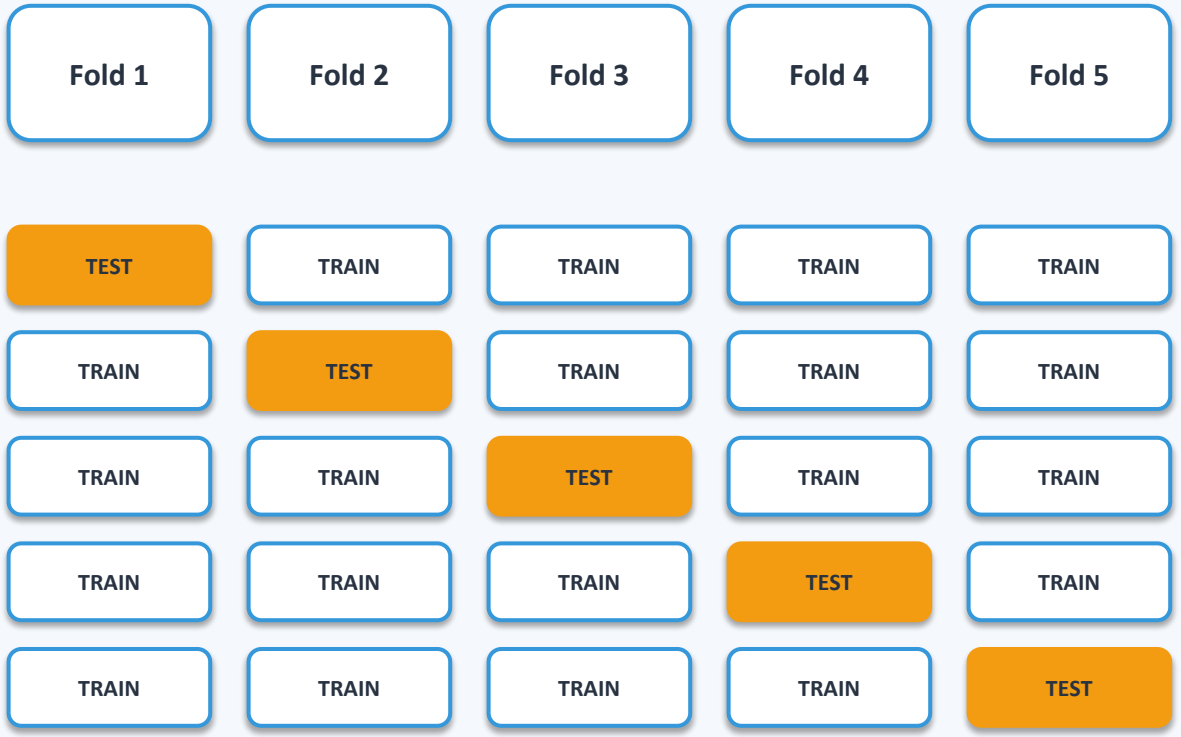
Holdout: one train/test split. Fast, but performance depends on the split.

Random subsampling: repeat random train/test splits and average the results.

Use stratification when class proportions must be preserved.

8. Cross Validation

K-fold CV rotates the test fold so every sample is evaluated



Average Validation Score

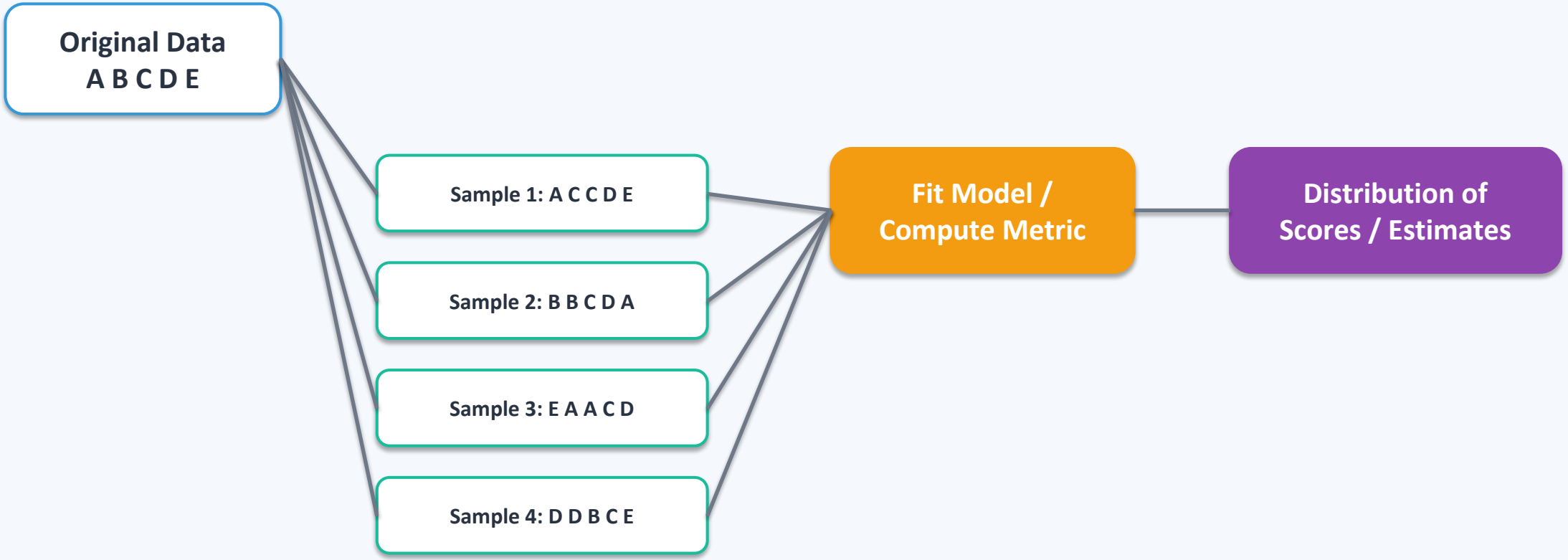
Typical choice: k = 5 or 10.

Stratified k-fold keeps class ratios similar across folds.

Nested CV can separate hyperparameter tuning from final evaluation.

9. Bootstrap

Estimate uncertainty by repeatedly sampling with replacement

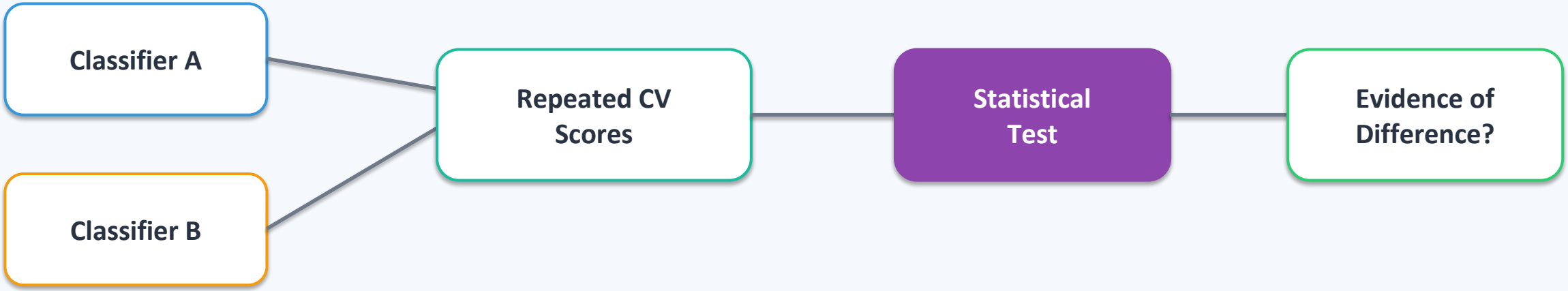


Sampling with replacement means observations may appear multiple times.

Useful for estimating variability, confidence intervals and model stability.

10. Model Selection Using Statistical Tests of Significance

A small score difference may be due to sampling variation



Null hypothesis H_o: no meaningful difference between classifiers.

Possible tests depend on the evaluation design: paired tests are often appropriate for paired CV results.

Report effect size and confidence intervals where possible—not only a p-value.

11. Comparing Classifiers: Cost–Benefit Analysis

The 'best' classifier depends on the cost of different errors

Outcome	Example	Cost
TP	Correct alert	Low
FP	False alarm	Medium
FN	Missed positive	High
TN	Correct rejection	Low

Expected Cost = Σ P(outcome) × Cost(outcome)

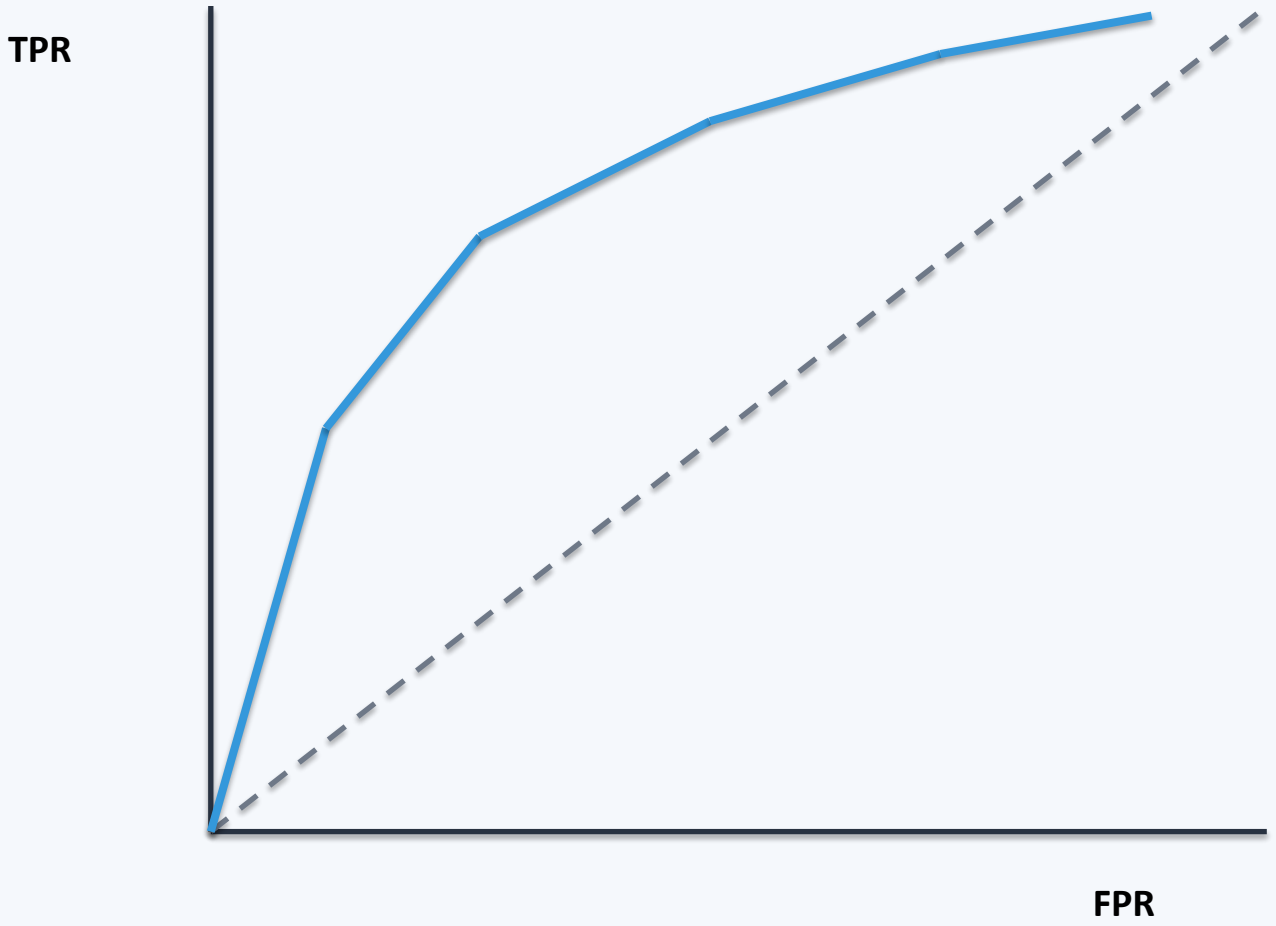
Medical screening: a false negative may be much more costly than a false positive.

Fraud detection: investigation cost matters when many legitimate transactions are flagged.

Choose thresholds using the real business/application costs.

12. ROC Curves

Visualize the trade-off between True Positive Rate and False Positive Rate



AUC = Area Under ROC Curve

TPR / Recall = TP ÷ (TP + FN)

FPR = FP ÷ (FP + TN)

Each threshold gives a different operating point.

AUC summarizes ranking ability across thresholds.

13. Choosing Evaluation Metrics

Match the metric to the problem and class distribution

Balanced classes

Accuracy can be informative

Accuracy / F1

Imbalanced classes

Minority class is important

Recall / Precision / F1 / PR-AUC

Ranking scores

Threshold changes

ROC-AUC / PR-AUC

Unequal error costs

FP and FN have different costs

Cost / expected loss

14. End-to-End Example

Fraud detection: combine ensembles + validation + cost-sensitive evaluation



Dataset: many legitimate transactions, few fraudulent ones.

Models: Random Forest and Gradient Boosting.

Evaluation: stratified CV + precision/recall/F1 + PR-AUC.

Decision: choose a threshold using the relative costs of missed fraud and false alarms.

KEY TAKEAWAYS

Ensembles combine multiple learners to improve generalization.

Bagging mainly reduces variance; Boosting builds learners sequentially.

Random Forest = bagged trees + random feature selection.

Imbalanced data requires metrics and training strategies beyond raw accuracy.

Holdout is simple; cross-validation gives a more robust estimate.

Bootstrap helps quantify variability and uncertainty.

Statistical tests help judge whether observed differences are meaningful.

ROC and cost–benefit analysis connect model scores to real decisions.

SELF-LEARNING → Explore XGBoost, LightGBM, SMOTE, PR Curves

Self-Learning Topics

Suggested topics for independent exploration / mini-seminar

Advanced Boosting
AdaBoost, Gradient Boosting, XGBoost, LightGBM

Imbalanced Learning
SMOTE, ADASYN, class weighting, threshold moving

Advanced Metrics
PR curve, MCC, balanced accuracy, calibration

Ensemble Diversity
Hard/soft voting, stacking, blending

Uncertainty
Bootstrap confidence intervals, calibration

Model Explainability
Feature importance, permutation importance, SHAP

Suggested activity: compare 3 classifiers on an imbalanced dataset and justify the metric + threshold.